# Krishna Mittal

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## Professional Summary

B.Tech CSE (AI & ML) student with hands-on experience in developing AI/ML and full-stack applications using Python, FastAPI, React, and modern ML frameworks. Experienced in RAG, LLMs, deep learning, NLP, and end-to-end AI pipelines, with a strong foundation in DSA, OOP, DBMS, and problem-solving.

## Education

**University of Petroleum and Energy Studies, Dehradun** — Aug 2023 - May 2027
Bachelor of Technology in Computer Science (AIML)
CGPA: 7.5/10

**Sri Chaitanya Techno School, Dehradun** — 2022 - 2023
CBSE Class-XII
Percentage: 65.6%

**Gautam International School, Dehradun** — 2021 - 2022
CBSE Class-X
Percentage: 84%

## Technical Skills

Programming Languages - Python, C++, Java, SQL
AI/ML - Machine Learning, Deep Learning, NLP, RAG, LLMs, CNN
Frameworks/Libraries - PyTorch, TensorFlow/Keras, scikit-learn, Sentence Transformers, FAISS, OpenCV
Web & Backend - FastAPI, React, HTML, CSS, REST APIs
Databases & Tools - MySQL, MongoDB, Git, GitHub, Jupyter, VS Code
Core CS - Data Structures & Algorithms, OOP, DBMS, Operating Systems

## Experience

**Survey of India, Dehradun** — 4 weeks (July 2025)
• Developed end-to-end Python GUI applications that automated critical workflows for the Geomagnetism Division.
• Collaborated with geophysicists and technical officers to understand domain requirements and convert them into functional software tools.

**IBM Internship** — 8 weeks (June 2026 - July 2026)
• Developed an AI-powered stock market analytics platform combining FinBERT financial sentiment, technical indicators, and machine-learning price-direction prediction for US and Indian equities.
• Built an 18-feature ML pipeline using XGBoost, LightGBM and Random Forest, and integrated FastAPI, Finnhub, yfinance, Reddit feeds and explainable BUY/HOLD/SELL recommendations.

## Projects

### AI Multimodal Interview Preparation Platform
Python, FastAPI, PyTorch, React, Scikit-learn
• Built a full-stack AI interview platform (FastAPI + React) simulating real-world interview scenarios using LLM-driven dynamic question generation.
• Designed resume-aware, adaptive follow-up questioning to personalize each interview session.
• Implemented real-time evaluation streaming and speaker-adaptive baseline calibration to improve scoring accuracy and personalization.

### AI-Powered Smart Courtroom - Legal RAG System
Python, FastAPI, React, FAISS, Sentence Transformers, Groq LLM
• Built a Retrieval-Augmented Generation (RAG) system to answer legal-case queries using judgment passages with grounded citations.
• Implemented hybrid legal retrieval using FAISS + BM25, Reciprocal Rank Fusion, cross-encoder reranking, metadata-aware chunking, and citation/evidence verification.
• Developed multi-agent Moot Court simulation, private case workspaces, bilingual Citizen Legal Advisor, BNS-IPC transition analysis, and PDF/image notice scanning with OCR.

### Plant Disease Detection System
Python, TensorFlow/Keras, NumPy, Matplotlib
• Developed an intelligent plant disease detection system using deep learning to assist in early identification of crop diseases and reduce agricultural losses.
• Designed and trained a Convolutional Neural Network (CNN) model for multi-class image classification of plant leaf diseases.
• Applied image preprocessing and augmentation techniques to improve model robustness under real-world conditions.